

Deep Learning for Solving Dynamic Models

Session 9: Continuous-time heterogeneous agents: HJB, KFE, and PINNs

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Based on: Achdou, Han, Lasry, Lions & Moll (2022), *Rev. Econ. Stud.* 89(1), 45–86.

Gu, Laurière, Merkel & Payne (2024), “Global Solutions to Master Equations.” arXiv:2406.13726.

Roadmap of This Session (40 min)

1. **Why continuous time:** what the reformulation buys us (5 min)
2. **The HJB–KFE system:** individual optimization plus the cross-sectional distribution; Aiyagari mapped operator by operator (20 min)
3. **Numerics:** upwind finite differences versus a PINN on the identical problem, and when each stops being viable (15 min)

Where this sits

Session 8 turned a PDE residual into a loss. Here that machinery meets the canonical heterogeneous-agent model, and we compare it against the classical finite-difference benchmark.

Why Continuous Time?

Advantages of continuous-time formulation:

- ▶ **Analytical tractability:** Ito calculus provides closed-form FOCs
- ▶ **Clean separation:** HJB (backward) $\leftrightarrow$ KFE (forward)
- ▶ **Powerful numerical methods:** finite differences, PINNs
- ▶ **No expectations operator:** replaced by differential operators
- ▶ **Connection to mean field games** (Lasry & Lions, 2007)

Key References

- ▶ Achdou, Han, Lasry, Lions & Moll (2022), *REStud*
- ▶ Kaplan, Moll & Violante (2018), *AER*
- ▶ Moll lecture notes:
benjaminmoll.com/lectures/

Discrete $\rightarrow$ Continuous

Day 4 (DEQNs, Young's method) used *discrete time*.

Today: *continuous time* $\Rightarrow$ PDEs.

The Aiyagari–Bewley–Huggett Framework

- ▶ Benchmark models in “quantitative” macroeconomics:
 - ▶ **Bewley** (1980, 1983): precautionary saving with borrowing constraints
 - ▶ **Huggett** (1993, *JEDC*): endowment economy, risk-free rate determination
 - ▶ **Aiyagari** (1994, *QJE*): production economy, general equilibrium
 - ▶ **Krusell & Smith** (1998, *JPE*): adds aggregate TFP shocks
- ▶ Core mechanism: **idiosyncratic labor income risk** is **uninsurable**
 - ▶ Households want to borrow when income is low → borrowing limits prevent this
 - ▶ Instead, households accumulate assets as **precautionary savings**
- ▶ These are building blocks for **HANK** models (Kaplan, Moll & Violante, 2018)
 - ▶ Heterogeneous Agent New Keynesian: monetary/fiscal policy with inequality

Itô Processes and SDEs

General Itô Process

A stochastic process X_t follows an **Itô process** if:

$$dX_t = \mu(X_t, t) dt + \sigma(X_t, t) dB_t$$

- ▶ $\mu(X_t, t)$: **drift** (deterministic trend)
- ▶ $\sigma(X_t, t)$: **diffusion coefficient** (volatility / noise intensity)
- ▶ B_t : standard Brownian motion

Important examples:

Geometric Brownian Motion (GBM)

$$dS_t = \mu S_t dt + \sigma S_t dB_t$$

Used for: stock prices, GDP

Ornstein–Uhlenbeck (OU) Process

$$dZ_t = \eta(\bar{Z} - Z_t) dt + \sigma dB_t$$

Used for: productivity, interest rates

What Is the Kolmogorov Forward Equation?

The **Kolmogorov forward equation** (KFE), also known as the **Fokker–Planck equation**, describes how the **probability density** of a stochastic process evolves over time.

Origin: physics and applied mathematics.

- ▶ **A.N. Kolmogorov** (1931): general theory of continuous Markov processes
- ▶ **A.D. Fokker** (1914) & **M. Planck** (1917): diffusion of particles
- ▶ Applications: heat conduction, fluid dynamics, plasma physics, chemical kinetics, population genetics, neuroscience, ...

Key idea: micro dynamics (drift + noise) $\Rightarrow$ macro density law of motion.



A.N. Kolmogorov, Wikimedia Commons

The KFE: General Itô Diffusion

For $dX_t = \mu(X_t) dt + \sigma(X_t) dB_t$ with **state-dependent** drift and diffusion, the density $g(x, t)$ satisfies the **general KFE**:

Kolmogorov Forward / Fokker–Planck Equation

$$\frac{\partial g}{\partial t}(x, t) = -\frac{\partial}{\partial x} [\mu(x) g(x, t)] + \frac{1}{2} \frac{\partial^2}{\partial x^2} [\sigma(x)^2 g(x, t)]$$

Important: when σ depends on x , it enters *inside* the second derivative:

$$\frac{1}{2} \partial_{xx} [\sigma(x)^2 g] \neq \frac{\sigma(x)^2}{2} \partial_{xx} g$$

In **divergence form**: $\partial_t g = -\partial_x \underbrace{\left[\mu(x) g - \frac{1}{2} \partial_x (\sigma^2 g) \right]}_{=: \text{probability flux } J(x, t)}$ Conservation law: $\partial_t g + \partial_x J = 0$.

KFE in Economics: The Wealth Distribution

Setup: Agent i has wealth a_t^i and income state $n_t^i \in \{n_1, n_2\}$.

Individual wealth evolves as:

$$da_t^i = \underbrace{s(a_t^i, n_t^i)}_{\text{savings} = \text{income} - \text{consumption}} dt, \quad a_t^i \geq \underline{a}$$

Income n_t^i switches between $n_1 \leftrightarrow n_2$ with Poisson intensities λ_1, λ_2 .

The **cross-sectional density** $g_t(a, n)$ satisfies:

$$\partial_t g_t(a, n) = -\partial_a [s(a, n) g_t(a, n)] - \lambda(n) g_t(a, n) + \lambda(\hat{n}) g_t(a, \hat{n})$$

where $\hat{n}$ denotes the complementary income state.

- ▶ $-\partial_a [s \cdot g]$: flow of agents along the wealth axis (savings)
- ▶ $-\lambda(n) g(a, n)$: agents **leaving** state n (exit from $n \rightarrow \hat{n}$)
- ▶ $+\lambda(\hat{n}) g(a, \hat{n})$: agents **entering** state n (entry from $\hat{n} \rightarrow n$)

Stationary Wealth Distribution

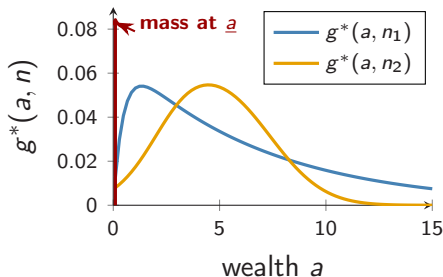
Setting $\partial_t g = 0$, the **stationary** (ergodic) distribution solves:

$$0 = -\partial_a [s(a, n) g^*(a, n)] \\ - \lambda(n) g^*(a, n) + \lambda(\hat{n}) g^*(a, \hat{n})$$

Key features of $g^*(a, n)$:

- ▶ **Right-skewed**: many poor, few rich
- ▶ **Mass point** at borrowing constraint $\underline{a}$ for low-income agents
- ▶ Separate densities for each income state
- ▶ Normalization: $\sum_n \int_{\underline{a}}^{\infty} g^*(a, n) da = 1$

This is the object solved numerically in Aiyagari (1994) and Achdou et al. (2022).



Deriving the HJB: Five Steps

Step 1: Dynamic Programming Principle. For small $h > 0$:

$$V(a, n) = \max_c \left\{ \int_0^h e^{-\rho t} u(c_t) dt + e^{-\rho h} \mathbb{E}[V(a_h, n_h) \mid a_0 = a, n_0 = n] \right\}$$

Step 2: Itô's lemma on $V(a_t, n_t)$ (between jumps of n):

$$dV = V'(a, n) \cdot s(c, a, n) dt$$

(no diffusion term since $da = s dt$ is deterministic between jumps)

Step 3: Take expectations, account for Poisson jumps:

$$\mathbb{E}[dV] = [V'(a, n) \cdot s + \lambda(n)(V(a, \hat{n}) - V(a, n))] dt$$

Step 4: Substitute into DPP, divide by h , let $h \rightarrow 0$:

Step 5: Impose optimality over c :

$$\rho V(a, n) = \max_c \{ u(c) + V'(a, n) \cdot (wn + ra - c) + \lambda(n)(V(a, \hat{n}) - V(a, n)) \}$$

The HJB Equation: Interpretation

$$\rho V(a, n) = \max_c \left\{ u(c) + \underbrace{V'(a, n) \cdot s(a, n, c)}_{\text{marginal value} \times \text{savings}} + \underbrace{\lambda(n)(V(a, \hat{n}) - V(a, n))}_{\text{expected change from income switch}} \right\}$$

Interpretation (“required return on the value function”):

- ▶ **LHS:** ρV = “required return” = discount rate $\times$ asset value
- ▶ **RHS:** “dividend” + “capital gain” = total return
 - ▶ $u(c)$: flow utility (“dividend”)
 - ▶ $V'(a, n) \cdot s$: expected capital gain from saving
 - ▶ $\lambda(n)(V(\hat{n}) - V(n))$: expected gain/loss from income switch

The HJB is an **asset pricing equation**:
the value function is priced like a financial asset.

Optimal Policy: The Consumption Function

First-order condition (differentiate HJB w.r.t. c):

$$u'(c^*) = V'(a, n) \quad \Longleftrightarrow \quad c^*(a, n) = (u')^{-1}(V'(a, n))$$

For CRRA utility $u(c) = \frac{c^{1-\gamma}}{1-\gamma}$ with $u'(c) = c^{-\gamma}$:

$$c^*(a, n) = [V'(a, n)]^{-1/\gamma}$$

Savings function:

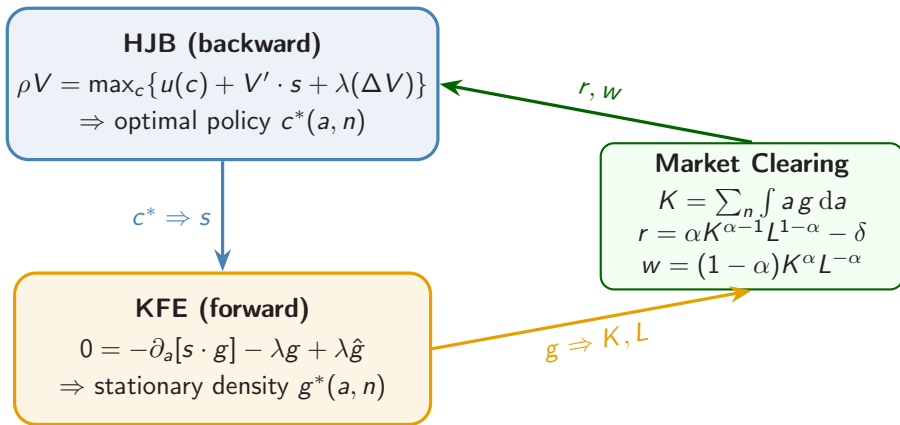
$$s(a, n) = wn + ra - c^*(a, n)$$

State constraint at $a = \underline{a}$:

$$\begin{aligned} 0 &< c^*(\underline{a}, n) \leq wn + r\underline{a}, \\ c^*(\underline{a}, n) &= \min\{[V_a(\underline{a}, n)]^{-1/\gamma}, wn + r\underline{a}\}, \\ s(\underline{a}, n) &= wn + r\underline{a} - c^*(\underline{a}, n) \geq 0. \end{aligned}$$

Interior: $u'(c^*) = V_a$. Boundary: constrain consumption so wealth cannot drift below $\underline{a}$.

The Coupled HJB–KFE System



Solution method: iterate until convergence (Aiyagari, no aggregate shocks):

1. Guess $r \rightarrow$ solve HJB for $V, c^* \rightarrow$ solve KFE for $g^* \rightarrow$ compute $K \rightarrow$ update r

Aiyagari (1994): Formal Stationary PDE System

Production side (no aggregate shocks):

$$Y = K^\alpha L^{1-\alpha}, \quad r = \alpha K^{\alpha-1} L^{1-\alpha} - \delta, \quad w = (1 - \alpha) K^\alpha L^{-\alpha}.$$

Household problem:

$$\max_{\{c_t\}} \mathbb{E} \left[\int_0^\infty e^{-\rho t} u(c_t) dt \right], \quad da_t = (wz_t + ra_t - c_t) dt, \quad a_t \geq \underline{a}, \quad z_t \in \{z_1, z_2\}.$$

HJB:

$$\begin{aligned} \rho V(a, z) &= \max_c \{ u(c) + V_a(a, z) [wz + ra - c] + \lambda(z) (V(a, \hat{z}) - V(a, z)) \}, \\ u'(c^*) &= V_a, \quad s(a, z) := wz + ra - c^*(a, z). \end{aligned}$$

KFE on $(\underline{a}, \infty)$:

$$0 = -\partial_a [s(a, z) g^*(a, z)] - \lambda(z) g^*(a, z) + \lambda(\hat{z}) g^*(a, \hat{z}).$$

Market clearing & normalization:

$$K = \sum_z \int_{\underline{a}}^\infty a g^*(a, z) da, \quad L = \sum_z \int_{\underline{a}}^\infty z g^*(a, z) da, \quad \sum_z \int_{\underline{a}}^\infty g^*(a, z) da = 1.$$

Unknowns: (V, g^*, r, w, K) (or (V, g^*, r) with $w = w(r)$).

Huggett vs. Aiyagari: Key Differences

	Huggett (1993)	Aiyagari (1994)
Economy	Endowment	Production ($Y = K^\alpha L^{1-\alpha}$)
Asset	Bond b	Capital claim a
Supply	Zero net ($\int b = 0$)	Positive ($\int a = K > 0$)
Price	Bond return q	Interest rate r , wage w
Clearing	$\int b g db = 0$	$\int a g da = K(r)$

How the distributions differ:

Huggett:

- ▶ Wealth centered around 0 (zero net supply)
- ▶ Large mass at borrowing constraint
- ▶ Symmetric-ish with constrained left tail

Aiyagari:

- ▶ Wealth shifted right ($K > 0$)
- ▶ Long right tail (wealthy agents)
- ▶ Smaller mass at constraint

The Master Equation

Key idea: Substitute KFE, market clearing, and belief consistency **into** the HJBE. Result: **one PDE** characterizing the equilibrium value function:

Master Equation

$$\begin{aligned} 0 = & -\rho V(a, n, z, g) + u(c^*) + V_a [w(z, g)n + r(z, g)a - c^*] \\ & + \lambda(n)(V(a, \hat{n}, z, g) - V(a, n, z, g)) + V_z \mu^z(z) + \frac{1}{2}(\sigma^z)^2 V_{zz} \\ & + \sum_{j \in \{1, 2\}} \int_{\underline{a}}^{\infty} \frac{\delta V}{\delta g_j}(a, n, z, g)(y) \mu_j^g(y, z, g) dy =: \mathcal{L}V. \end{aligned}$$

where c^* satisfies $u'(c^*) = \partial_a V(a, n, z, g)$.

“One equation to rule them all”

HJB + KFE + market clearing = master equation

Finite Differences: The Classical Approach

How have economists traditionally solved these models?

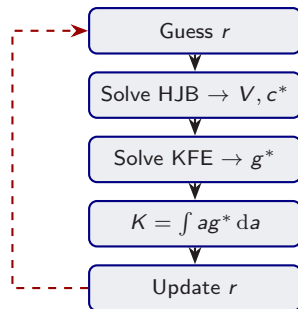
Finite difference method (Achdou et al., 2022):

1. Discretize wealth a on a grid $\{a_1, \dots, a_I\}$
2. Approximate derivatives by finite differences:

$$V'(a_i) \approx \frac{V_{i+1} - V_i}{\Delta a} \quad \text{or} \quad \frac{V_i - V_{i-1}}{\Delta a}$$

3. **Upwind scheme**: use forward diff. when savings > 0 , backward when < 0
4. Iterate HJB $\rightarrow$ policy $\rightarrow$ KFE $\rightarrow$ aggregates $\rightarrow$ prices $\rightarrow$ HJB

This summarizes the finite-difference workflow; code examples are provided later in Python.



Finite Differences: What They Produce

Finite-Difference Example: Aiyagari GE

The classical FD approach solves:

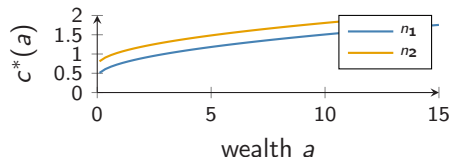
- ▶ HJB block: upwind discretization + implicit updates
- ▶ KFE block: transition operator + stationary distribution
- ▶ Price block: root-finding for market-clearing r

Produces:

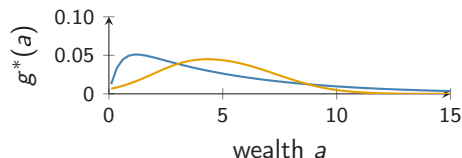
- ▶ Value function $V(a, n)$
- ▶ Consumption/savings policy $c^*(a, n)$
- ▶ Stationary density $g^*(a, n)$
- ▶ Equilibrium prices r^*, w^*

This is our ground truth for validating deep learning methods.

Consumption (FD solution)



Stationary density (FD solution)



Why Finite Differences Are Not Enough

FD works well for:

- ✓ Stationary HJB–KFE systems
- ✓ Low-dimensional states (a, n)
- ✓ Monotone convergence theory
- ✓ Fast sparse linear algebra

FD is strained by:

- ✗ Global KS master equation
- ✗ State (a, n, z, g) with distribution g
- ✗ Dense tensor-product grids
- ✗ Moment truncation of g

Takeaway

Neural methods do not remove the equilibrium conditions; they approximate value and distribution maps without building a full grid over g .

From Single PDE to Coupled System

Block 1 (PINNs): Solve a *single* PDE (Poisson, HJB cake-eating, Black–Scholes).

Block 2 (this block): Solve a *coupled system* of PDEs arising from equilibrium:

Component	PDE	Unknown
HJB (backward)	$\rho V = \max_c \{u(c) + V' \cdot s + \lambda(\Delta V)\}$	Value $V(a, n)$
KFE (forward)	$0 = -\partial_a[s \cdot g] - \lambda g + \lambda \hat{g}$	Density $g(a, n)$
Market clearing	$K = \sum_n \int a g \, da$	Prices r, w

The coupling: HJB output ($c^* \rightarrow s$) is the KFE input (drift).

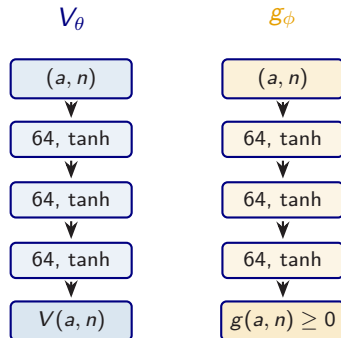
KFE output ($g \rightarrow K, L$) determines prices (r, w) that enter HJB.

PINN approach: Parameterize *both* V and g by neural networks.
Minimize *all* residuals **jointly** via a single loss function.

PINN Architecture for Aiyagari/Huggett

Two neural networks:

- ▶ $V_\theta(a, n)$: value function network
 - ▶ Input: (a, n) (n encoded as 0/1)
 - ▶ Output: scalar V
 - ▶ Architecture: 3–4 hidden layers, 64 neurons, tanh
- ▶ $g_\phi(a, n)$: density network
 - ▶ Input: (a, n)
 - ▶ Output: scalar $g \geq 0$ (use softplus)
 - ▶ Same architecture



Derived quantities (via auto-diff):

- ▶ $V'_\theta = \partial_a V_\theta \Rightarrow$ consumption: $c^* = (V'_\theta)^{-1/\gamma}$
- ▶ $\partial_a g_\phi \Rightarrow$ KFE residual

Training Algorithm: Jointly Solving HJB + KFE

PINN Training (Stationary Equilibrium)

- 1: Initialize V_θ , g_ϕ , guess $r^{(0)}$ (*FD reserved for validation*)
- 2: **for** $k = 1, \dots, N_{\text{epochs}}$ **do**
- 3: Sample M collocation points $\{(a_m, n_m)\}$
- 4: Compute $c_m^* = (V'_\theta(a_m))^{-1/\gamma}$
- 5: Compute R_{HJB}^m , R_{KFE}^m via auto-diff
- 6: Integrate KFE on a fixed asset grid
- 7: Compute $\mathcal{L}_{\text{SC}}, \mathcal{L}_{\text{flux}}, \mathcal{L}_{\text{mass}}, \mathcal{L}_{\text{MC}}$
- 8: Total loss: $\mathcal{L} = \sum \kappa_i \mathcal{L}_i$; update θ, ϕ (Adam)
- 9: **if** $k \bmod N_r = 0$ **then**
- 10: $r \leftarrow r + \eta(K^s - K^d)$
- 11: **end if**
- 12: **end for**
- 13: Freeze collocation grid; run FP64 L-BFGS polish

Key choices:

- **Collocation:** sample (a, n) uniformly on $[\underline{a}, \bar{a}] \times \{n_1, n_2\}$
- **Price update:** update r slowly (every N_r steps) to let NN adapt
- **Loss weights:** start with $\kappa_{\text{HJB}} = \kappa_{\text{KFE}} = 1$, larger $\kappa_{\text{SC}}, \kappa_{\text{flux}}, \kappa_{\text{MC}}$
- **Optimizer:** Adam, $\text{lr} \sim 10^{-3}$, then FP64 L-BFGS
- **Validation:** compare against FD only after training

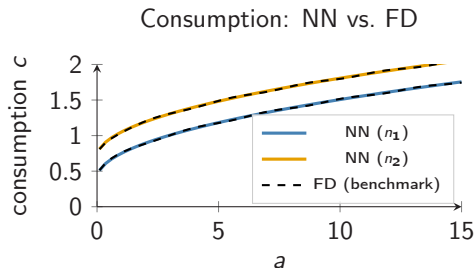
Same philosophy as Day 5: **economic equations**
= **loss function**; **auto-diff** = **all derivatives**.

Results: Aiyagari Model (No Aggregate Shocks)

Validation: Compare EMINN solutions against finite-difference benchmark.

Method	ME Loss	MSE(NN,FD)
Finite Pop. NN	3.1×10^{-5}	4.8×10^{-5}
Discrete State NN	9.3×10^{-6}	6.6×10^{-5}

- ▶ ME residual: $\sim 10^{-5}$ (excellent)
- ▶ Agreement with FD: $\sim 10^{-5}$ MSE
- ▶ Both approaches match FD closely
- ▶ Consistent across multiple runs



Why PINNs over Finite Differences?

For stationary problems (Huggett/Aiyagari without aggregate shocks):

	Finite Differences	PINN
Grid required	Yes (I points)	No (collocation)
Upwind scheme	Manual implementation	Auto-diff handles it
Curse of dimensionality	$O(I^d)$	$O(M \cdot \theta)$
Extension to higher dim	Very hard	Natural
Borrowing constraint	Special treatment	Soft penalty / hard BC
Accuracy (2D)	$\sim 10^{-6}$	$\sim 10^{-4}$
Speed (2D)	Faster	Slower

Verdict: *FD is faster and more accurate* for low-dim stationary problems, use it as the gold-standard benchmark. PINNs **scale to high dimensions**: aggregate shocks (KS, state $\ni g$), multiple assets, richer heterogeneity.

$\Rightarrow$ Part III: **EMINNs** for the master equation.

Method Comparison: FD vs. PINN vs. EMINN

	Finite Diff.	PINN	EMINN
Global master eq.	✗	✗	✓
Stationary equilibrium	✓	✓	✓
Grid required	Yes	No	No
High-dimensional scaling	Limited	Better, validate	Better, validate
Handles g as state	Standard: no	No	Via $\hat{\phi}$
Theory	Strong in low dim	Limited	Limited
Low-dim accuracy	$\sim 10^{-6}$	problem-dependent	reported $\sim 10^{-5}$

Bottom line:

- ▶ **Stationary, low-dim:** FD is fast and reliable. Use it as benchmark.
- ▶ **Aggregate shocks:** EMINNs target global master-equation solutions.
- ▶ **PINNs:** useful bridge for the coupled HJB–KFE system.

Code Companion: Run This Live

Day 6 companion notebooks (self-contained, PyTorch):

- ▶ **NB 06** (*self-study*): 06_PE_Discrete_HJB_PINN.ipynb, 1D PE with discrete income: upwind FD + PINN
- ▶ **NB 07** (*self-study*): 07_PE_Diffusion_HJB_PINN.ipynb, 2D PE with OU earnings: 2D FD + PINN
- ▶ **NB 08** (*in-class*): 08_Aiyagari_Continuous_Time_FD_and_PINN_PyTorch.ipynb, full GE Aiyagari: HJB + KFE + market clearing

Suggested in-class demo flow:

1. Start with NB 06 (simplest, self-study): FD convergence, then PINN overlay
2. Move to NB 07 (self-study): 2D contour plots, cross-sections, mesh-free advantage
3. Focus on NB 08 (in-class): full GE loop, PINN vs FD agreement diagnostics